THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** photosynthesis	Using sunlight.
2	**(c)** straight line	Straight.
3	**(c)** axis	Axis.
4	**(a)** transpiration	Evaporation.
5	**(d)** image	Reflection.
6	**(c)** hot	Warm.
7	**(a)** green	Colour.
8	**(b)** reflection	Equal.
9	**(c)** Sun and Earth	Blocks Sun.
10	**(b)** chlorophyll	Pigment.
11	**(a)** luminous object	Gives light.
12	**(d)** Sun	Sun.
13	**(b)** insects	Insectivorous.
14	**(a)** reflection	Reflection.
15	**(b)** cold	Cool.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Food making by plants (1); Needs sunlight (1).	1+1
17	Gives its own light (1); E.g. the Sun (1).	1+1
18	Spinning on an axis (1); Causes day and night (1).	1+1
19	Small pores in leaves (1); Gas exchange (1).	1+1
20	A line perpendicular to the surface (1); At the point of incidence (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Carbon dioxide + water (1); Sunlight and chlorophyll (1); Glucose + oxygen (1).	1+1+1
22	Sun (1); Candle (1); Bulb (1).	1+1+1
23	Day and night (1); Sunrise and sunset (1); Apparent motion of stars (1).	1+1+1
24	Anchor plant (1); Absorb water (1); Absorb minerals (1).	1+1+1
	OR: Erect (1); Same size (1); Virtual (1).	1+1+1
25	A star (1); Source of light (1); Hot (1).	1+1+1
	OR: Boil in alcohol (1); Add iodine (1); Blue-black colour (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Occurs in leaves (1); Needs light (1); Needs CO2 and water (1); Makes glucose (1); Gives oxygen (1).	1+1+1+1+1
	OR: Luminous give light (1); Non-luminous do not (1); Sun vs Moon (1); Seen by own light vs reflected (1); Examples (1).	1+1+1+1+1
27	Earth is a sphere (1); Spins on its axis (1); Faces the Sun: day (1); Away: night (1); 24 hours cycle (1).	1+1+1+1+1
	OR: Trap insects (1); Pitcher plant (1); Venus flytrap (1); Need nitrogen (1); Have green leaves (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** day	For CO_2.
2	**(a)** normal	Perpendicular.
3	**(d)** Sun and Moon	Shadow on Moon.
4	**(d)** stomata	Pores.
5	**(a)** non-luminous	Reflects light.
6	**(b)** Earth	Earth.
7	**(d)** parasite	Harms host.
8	**(b)** erect	Upright.
9	**(c)** longer	Longer.
10	**(d)** wastes	Excretion.
11	**(c)** incident ray	Falling.
12	**(c)** new moon	Aligned.
13	**(c)** roots	Root hairs.
14	**(c)** transparent	Clear.
15	**(c)** rotation	Rotation.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	366 days (1); Every 4 years (1).	1+1
17	Lives on and feeds on host (1); Cuscuta (1).	1+1
18	Glass (1); Water (1).	1+1
19	Moon blocks the Sun (1); At new moon (1).	1+1
20	Chlorophyll (1); Green (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	No light (1); Objects do not reflect (1); Need light to see (1).	1+1+1
22	24 hours (1); One rotation (1); Day and night (1).	1+1+1
23	Blade (1); Veins (1); Stalk (1).	1+1+1
24	Moon (1); Table (1); Book (1).	1+1+1
	OR: Year (1); Seasons (1); Changing constellations (1).	1+1+1
25	Support (1); Carry water (1); Carry food (1).	1+1+1
	OR: Angle of incidence = angle of reflection (1); Rays and normal in one plane (1); Both (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Three cards (1); Holes aligned (1); Candle on one side (1); See the light (1); Shift one card: no light (1).	1+1+1+1+1
	OR: Full moon (1); Earth between (1); Shadow on Moon (1); Moon looks dark red (1); Safe to watch (1).	1+1+1+1+1
27	Boil in water (1); Boil in alcohol (1); Wash (1); Add iodine (1); Note colour (1).	1+1+1+1+1
	OR: Take a lamp (1); Hold an opaque object (1); Place a screen (1); Shadow forms (1); Change distance (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon and the Sun** | Class 7 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** saprotroph	Dead matter.
2	**(c)** object	Same size.
3	**(d)** shorter	Shorter.
4	**(d)** day and night	Always.
5	**(c)** reflected ray	Returns.
6	**(a)** full moon	Aligned.
7	**(d)** sunlight	Light.
8	**(c)** opaque	Blocks light.
9	**(c)** tilt of Earth	Tilt.
10	**(a)** fungus	Together.
11	**(b)** right	Lateral inversion.
12	**(d)** 0° latitude	Middle.
13	**(c)** chloroplasts	Cell part.
14	**(d)** 20°	Equal.
15	**(c)** solar eclipse	Harmful.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Does not give light (1); E.g. the Moon (1).	1+1
17	Going round the Sun (1); Takes a year (1).	1+1
18	Loss of water vapour (1); From leaves (1).	1+1
19	A box with a tiny hole (1); Forms an image (1).	1+1
20	A natural satellite (1); Of the Earth (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Pinhole (1); Special glasses (1); Screen projection (1).	1+1+1
22	Fix nitrogen (1); Enrich soil (1); Better crops (1).	1+1+1
23	Blocks more rays (1); Larger shadow (1); Closer to the source (1).	1+1+1
24	365 days (1); One revolution (1); Plus quarter day (1).	1+1+1
	OR: Autotrophic (1); Parasitic (1); Saprotrophic (1).	1+1+1
25	Glass (1); Water (1); Air (1).	1+1+1
	OR: New moon (1); Half moon (1); Full moon (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Axis tilted (1); Earth goes round the Sun (1); Summer: tilted towards (1); Winter: tilted away (1); Change in weather (1).	1+1+1+1+1
	OR: Bacteria (1); In root nodules (1); Fix nitrogen (1); Legumes (1); Symbiosis (1).	1+1+1+1+1
27	Opaque object (1); Light source (1); Screen (1); Size changes (1); Dark (1).	1+1+1+1+1
	OR: Do not look at the Sun (1); Use pinhole (1); Use safe filters (1); Protect children (1); Follow experts (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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